LUKE KIMBALL

Software Engineer — Multi-Model Agent Harnesses · Sandboxed Runtimes · CLI & OpenRouter Tooling

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Systems and applied AI engineer maintaining 50+ open-source repositories (github.com/duketopceo) across Rust, Go, TypeScript, and Python, focused on multi-model agent harnesses, CLI workflow runners, masked credential brokering, context compaction, and OpenRouter evaluation tooling. Built laptop-to-cloud agent runtimes (Khan, open-computer), cross-harness CLI adapters (agentflow), and OS keyring isolation (OmaSeal) so sandboxed agents switch models and execute tools without ever seeing raw credentials.

TECHNICAL STACK & HARNESS FLUENCY

- **Languages & Systems:** Rust (Tokio, Axum, Serde, SQLite FTS5, sqlite-vec), Go (Chi, OS daemons, D-Bus), TypeScript / Node.js (Hono, Next.js, Drizzle), Python (FastAPI, LangGraph, asyncio), PostgreSQL 17 (RLS, skip-locked queues).
- **Agent Harnesses & Routing:** OpenRouter API & Management Keys, LiteLLM, Temporal & Restate durable workflows, Model Context Protocol (MCP), Claude Code, Codex, OpenCode, Cursor, Devin, Agent Zero, Hermes, Ollama, llama.cpp (Vulkan).
- **Sandboxing, Secrets & Quality:** E2B & CubeSandbox isolated runtimes, Freedesktop Secret Service (gnome-keyring), 1Password CLI, AES-256-GCM vaults, SSRF egress allowlists, Vision E2E testing (Argus), Docker Swarm, GitHub Actions CI.

SELECTED AGENT RUNTIME, CLI & OPENROUTER SYSTEMS

Khan & OpenRouter Usage Plugin — Agent Harness & Spend Telemetry — Python · OpenRouter · Temporal · LiteLLM · SQLite · MCP

- OpenRouter-native fork of Agent Zero with Temporal workflow durability, LiteLLM multi-model worker pools, MCP tool integration, dynamic tool risk scoring, and a one-command PATH CLI plus macOS app wrapper.
- Built openrouter_usage (and a0-openrouter-spend), the published Agent Zero OpenRouter Usage Plugin featuring server-side management-key isolation, workspace-scoped /analytics/query cost attribution, local SQLite usage caching, and confirmation-gated model routing.

agentflow & open-computer — Cross-Harness CLI & Sandboxed Cloud Runtime — TypeScript · Node.js · Python · OpenRouter · Temporal · CubeSandbox

- Engineered @lukedaduke/agentflow (v0.2.0 on npm), a cross-harness .mjs CLI workflow runner that orchestrates bounded non-interactive sessions across Claude Code, Codex, Cursor, Devin, and OpenCode with disk-journaled state recovery.
- Built open-computer with a benchmarked multi-model router (bench/router/), disposable CubeSandbox execution, a policy and PII approval gate, a portable /skills capability registry across models, and a /automations Temporal scheduler.

OmaSeal & a0-plugin-omaseal — OS Keyring & Masked Credential Brokering — Go · Python · gnome-keyring · 1Password · AES-256-GCM · MCP

- Built a Go desktop keyring engine (v0.4.0) over the Freedesktop Secret Service API and MCP, plus a companion Python secrets resolver across gnome-keyring, 1Password, and AES-256-GCM vaults.
- Substitutes §§secret(NAME) placeholders in-process at tool execution and scrubs resolved values from tool output so sandboxed agents only ever see secret names and masked values.

jev-compact & Kurultai — Context Compaction & Hybrid Retrieval Daemon — Rust · Tokio · Axum · Python · SQLite FTS5 · sqlite-vec · MCP

- Created jev-compact (and a0-plugin-jev-compact), an agent session-compaction engine that uses a decision model to score transcript spans, keep load-bearing context verbatim, tombstone stale segments, and restore spans on demand over MCP.
- Developed kurultai (v0.7.0, 342 commits) in async Rust to index notes, chat transcripts, and repositories into SQLite with two-hop tag FTS5 expansion, Reciprocal Rank Fusion (RRF) hybrid scoring, and deterministic graph_chain citation provenance over MCP.

orchestral & Argus — Multi-Model Evaluation & Vision E2E Harness — Python · TypeScript · Playwright · OpenRouter · E2B · Docker · CI

- Built orchestral (422 commits, 125 test files) and an E2B-compatible isolated code runtime to evaluate whether low-cost orchestrator and worker model pairs match frontier-model output quality across OpenRouter providers under CI gates.
- Shipped Argus (argus-reviewer-e2e v0.5.0 on npm, 302 commits), a BYOK OpenRouter vision-model E2E testing harness and PR reviewer with fingerprint-cached replay, DOM-verified grounding, hard per-run budget caps, and networkless Docker reproduction; dogfooded across tallyline.

Pace Server & tallyline — Multi-Tenant Cloud & Agent Dispatch Monorepo — Rust (Axum) · Go (Chi) · TypeScript (Hono) · PostgreSQL 17 · RLS · MCP

- Designed Pace's multi-tenant cloud backend (Rust/Axum gateway, Go/Chi services, Postgres 17 RLS, Stripe Connect metering, Stytch B2B SSO) with an 8-step agentrun pipeline behind TierGate autonomy controls and SSRF egress allowlists.
- Built tallyline (v1.0.0), an agent-native issue tracker monorepo with an MCP server, CLI, and single-dependency PostgreSQL skip-locked agent dispatch worker; isolated and reported an upstream wake-queue attribution race in paperclipai/paperclip (#14121).

dayflow-linux & portfolio-hub — Linux Desktop Journal & Telemetry — Go · TypeScript · Next.js · SQLite FTS5 · OpenRouter · MCP

- Ported Dayflow natively to Linux in Go (v1.6.0, 212 commits) with local-first screen capture, OpenRouter vision summarization, standup generation, agent-session recaps, FTS5 search, and MCP tool exposure (including 6 merged external community PRs).
- Built portfolio-hub (luke-the-duke.com) with cached 90-day GitHub App engineering telemetry, machine-readable JSON Resume and llms.txt endpoints, and automated privacy and visual regression gates across 50+ repositories.

PROFESSIONAL EXPERIENCE

IT Support & Data Specialist — Bartlett Roofing — 2025 – Present · Murray, UT (UT / ID)

- Promoted within 4 months to own internal systems, data automation, and multi-state IT infrastructure across Utah and Idaho for a 200-person commercial and residential roofing organization.
- Deployed containerized internal services (Snipe-IT for 105+ assets, MeshCentral, Vaultwarden, Docker, Tailscale) on bare-metal Linux alongside authenticated read/write APIs, Python permit automations, and a Next.js operations dashboard.
- Trained the engineering and IT team on daily Claude Code, Cursor, and multi-model OpenRouter workflows in month one.

Data Intern & Manufacturing IT Liaison — LiveView Technologies (LVT) — 2024 – 2025 · Orem, UT

- Automated production data-pipeline fixes in Python and built executive Tableau dashboards tracking defect rates, assembly throughput, and cycle times across manufacturing and engineering teams.

Inside Sales & Onboarding Trainer — Vivint Smart Home — 2020 – 2024 · Provo, UT

- Ranked top 6 in department for 4 consecutive months on commission-only compensation and trained 30+ new hires on technical product workflows.

EDUCATION

Certificate in Data Analytics — Mountainland Technical College (MTECH) — 2024 – 2025

Coursework (3 Semesters) in Computer Engineering & Economics — Brigham Young University-Idaho — 2018 – 2020